# Are Expensive Wines Necessarily Better-Rated?

Exploring the relationship between wine price and ratings using **data** from Vivino.com

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# Variables

## Independent Variable (X): Price (€) 💰

Price represents our **input** variable that we can control or observe. It's the factor we believe might influence wine ratings.

## Dependent Variable (Y): Ratings ⭐

Ratings represent our **output** variable that we measure in response to price changes. These are the numerical scores given to wines by consumers on Vivino.com.

By examining how ratings (Y) change in response to different price points (X), we investigate whether higher prices truly correlate with better wine ratings.

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# Multiple Possible Outcomes: Null vs. Alternative Hypothesis

- Null Hypothesis: There is no significant relationship between wine price and rating.
- Alternative Hypothesis: There is a significant relationship between wine price and rating.

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# Confounding Factors in Wine Analysis

Several external variables can influence the relationship between price and ratings:

### Country/Region

Regional reputation and territory

### Wine Type

Different styles like White, Red, Rosé

### Wine Age

Vintage vs. Newer

### Rating Volume

Popularity bias

### Packaging

Bottle design and labeling

### Brand

Brand recognition bias

### Season

Seasonal demand

### Winery Reputation

Historical prestige

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# The Ideal Experiment: Blind Taste Tests

This experimental design allows us to isolate the effect of confounding factors on wine price and ratings.

### Control Group

Wine ratings collected from participants who know the prices

### Treatment Group

Wine ratings collected from participants blind to price information

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# Descriptive Statistics

| | Name | Country | Region | Winery | Rating | NumberOfRatings | Price | Year | wine_type |
|---|---|---|---|---|---|---|---|---|---|
| count | 13834 | 13834 | 13834 | 13834 | 13834.000000 | 13834.000000 | 13834.000000 | 13834 | 13834 |
| unique | 10934 | 33 | 861 | 3505 | NaN | NaN | NaN | 34 | 4 |
| top | Chardonnay 2018 | Italy | Rioja | Markus Molitor | NaN | NaN | NaN | 2018 | Red |
| freq | 78 | 3919 | 385 | 73 | NaN | NaN | NaN | 2723 | 8666 |
| mean | NaN | NaN | NaN | NaN | 3.865664 | 428.322466 | 33.024850 | NaN | NaN |
| std | NaN | NaN | NaN | NaN | 0.296427 | 1838.413812 | 70.899893 | NaN | NaN |
| min | NaN | NaN | NaN | NaN | 2.200000 | 25.000000 | 3.150000 | NaN | NaN |
| 25% | NaN | NaN | NaN | NaN | 3.700000 | 56.000000 | 9.902500 | NaN | NaN |
| 50% | NaN | NaN | NaN | NaN | 3.900000 | 129.000000 | 15.950000 | NaN | NaN |
| 75% | NaN | NaN | NaN | NaN | 4.100000 | 336.000000 | 32.500000 | NaN | NaN |
| max | NaN | NaN | NaN | NaN | 4.900000 | 94287.000000 | 3410.790000 | NaN | NaN |

*80% of the data constitutes of wine from Europe | 63% of the data constitutes of Red Wine*

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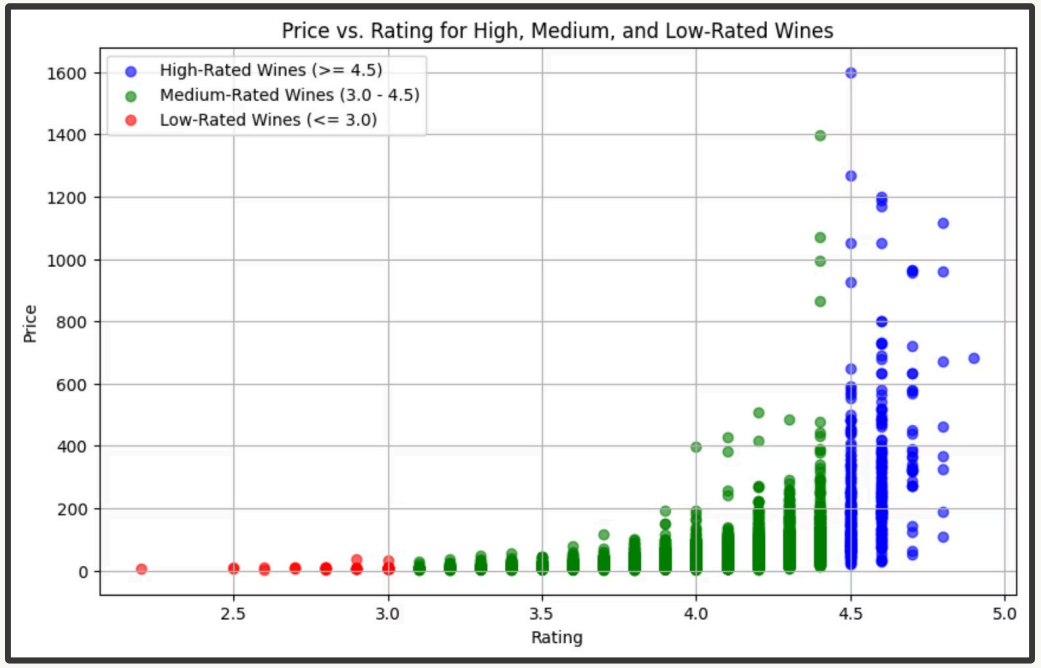
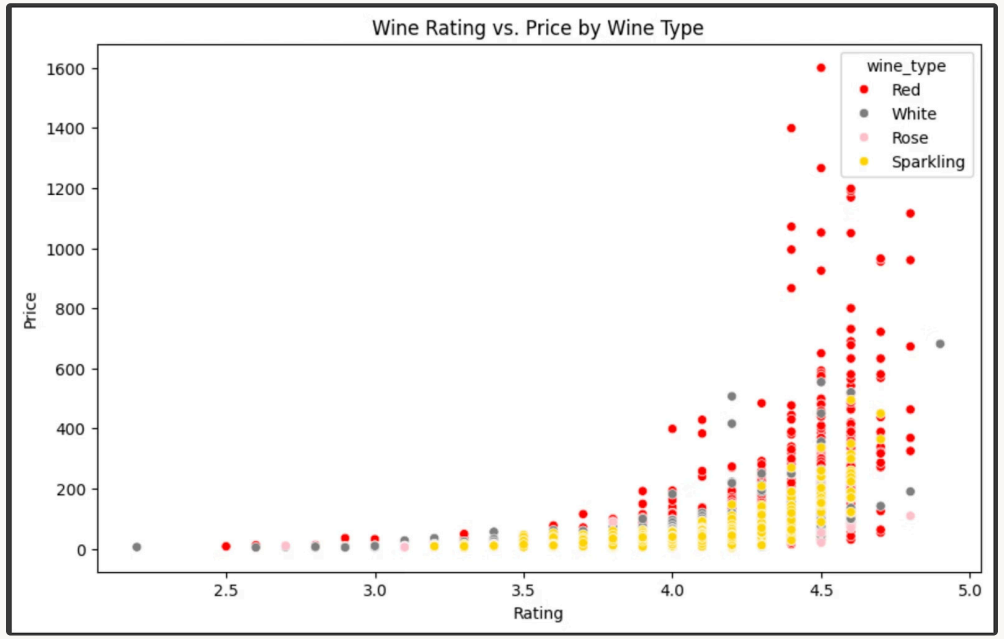
# Exploratory Data Analysis (1/3)

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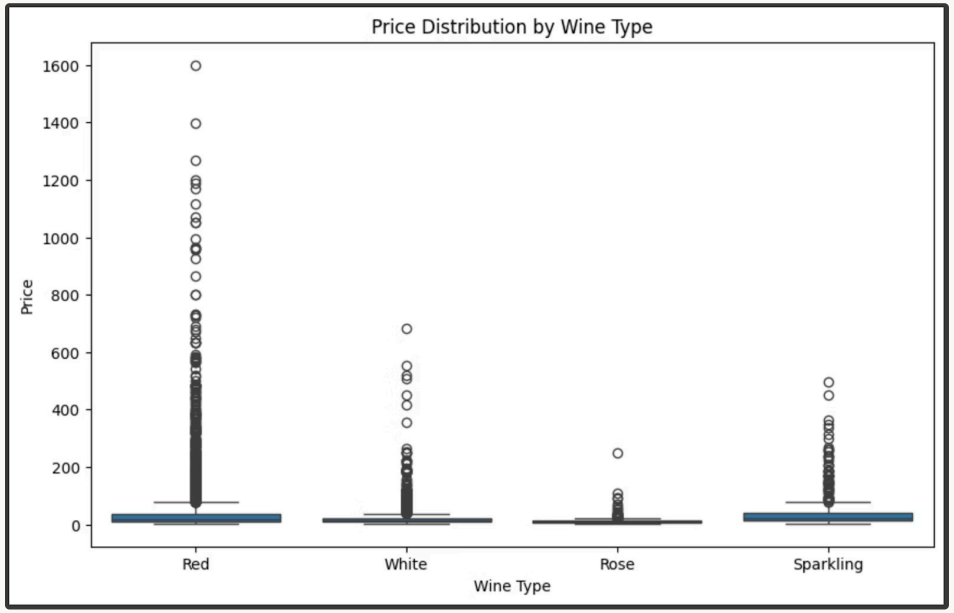
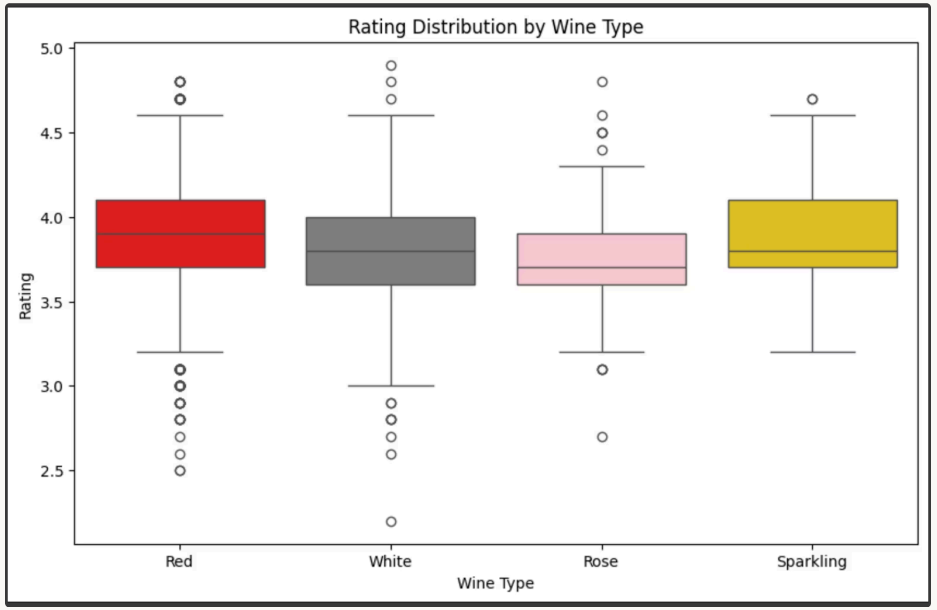
# Exploratory Data Analysis (2/3)

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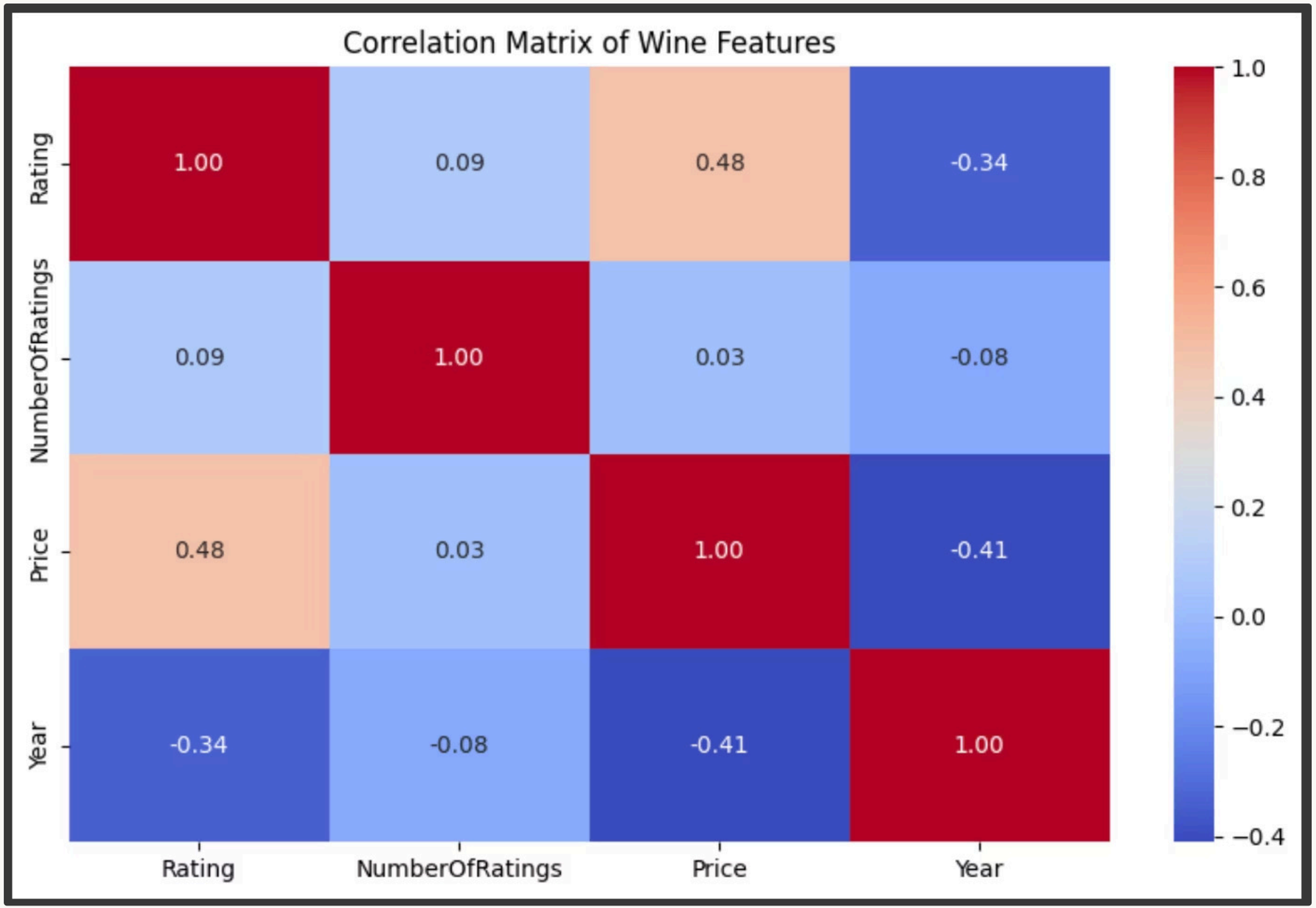
# Exploratory Data Analysis (3/3)

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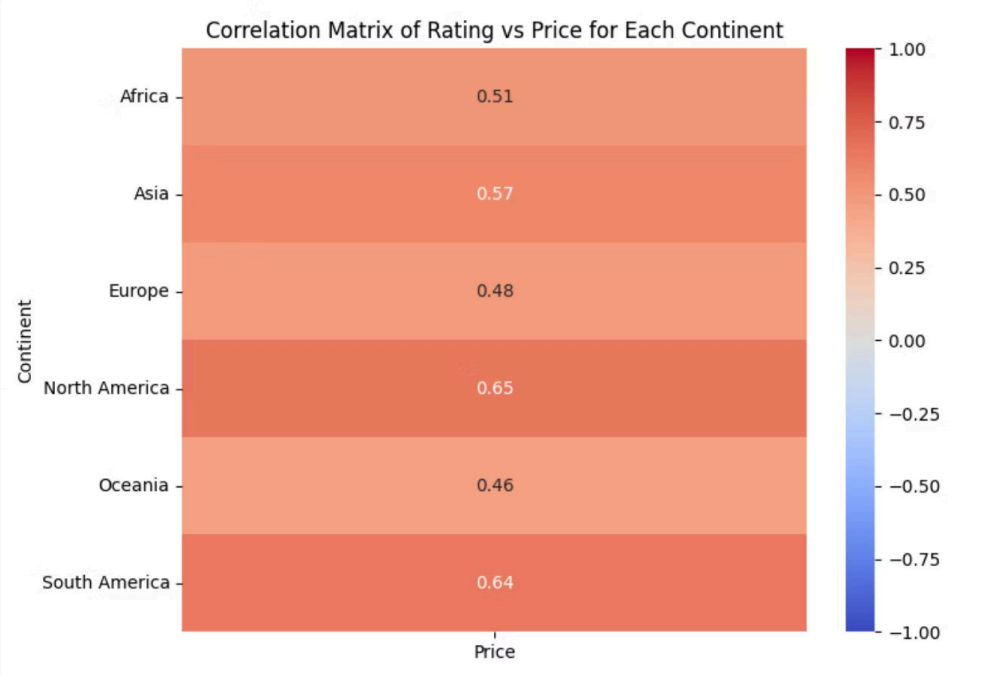
# SubTesting - Continent & Wine Type



Correlation Matrix Value of Rating vs Price is **0.48**

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# Simple Linear Regression Model

OLS Regression Results

| | | | |
|---|---|---|---|
| **Dep. Variable:** | Rating | **R-squared:** | 0.233 |
| **Model:** | OLS | **Adj. R-squared:** | 0.233 |
| **Method:** | Least Squares | **F-statistic:** | 4196. |
| **Date:** | Wed, 26 Feb 2025 | **Prob (F-statistic):** | 0.00 |
| **Time:** | 20:57:03 | **Log-Likelihood:** | -971.45 |
| **No. Observations:** | 13833 | **AIC:** | 1947. |
| **Df Residuals:** | 13831 | **BIC:** | 1962. |
| **Df Model:** | 1 | | |
| **Covariance Type:** | nonrobust | | |

| | coef | std err | t | P>\|t\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| **const** | 3.7933 | 0.002 | 1533.677 | 0.000 | 3.788 | 3.798 |
| **Price** | 0.0022 | 3.4e-05 | 64.774 | 0.000 | 0.002 | 0.002 |

| | | | |
|---|---|---|---|
| **Omnibus:** | 2011.746 | **Durbin-Watson:** | 1.984 |
| **Prob(Omnibus):** | 0.000 | **Jarque-Bera (JB):** | 7123.543 |
| **Skew:** | -0.718 | **Prob(JB):** | 0.00 |
| **Kurtosis:** | 6.209 | **Cond. No.** | 81.4 |

**Key Takeaways from the Regression Output:**

1. **R-squared (0.233)** → About **23.3%** of the variation in wine ratings is explained by price.
   - This suggests that price has a **weak** influence on rating—other factors likely have a greater impact.
2. **Coefficient for Price (0.0022)** → Positive but very small.
   - For every **$1 increase** in price, the rating increases by **0.0022 points**—which is **not practically significant**.
3. **p-value (0.000)** → Statistically significant relationship.
   - The effect of price on ratings is **real (not due to random chance)**, but **the impact is weak**.

**Conclusion**

- **Higher-priced wines tend to have slightly higher ratings**, but the effect is **very weak**.
- **Price is NOT a strong predictor of wine ratings**—other factors (like winery, region, or grape variety) likely influence ratings.

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# Summary of Analysis

## 1 Similar Average Rating Across Wine Types

Despite differences in price range and median, all four types of wine have a similar average rating.

## 2 Price is Not a Strong Predictor

Regression analysis indicates that price is NOT a strong predictor of wine ratings.

## 3 Potential Non-Linear Relationship

While there isn't a strong linear relationship, the pattern suggests a different relationship between wine prices and ratings may exist.

## 4 Positive Correlation Across Regions

A positive correlation exists between price and rating across all continents and wine types, with the strongest correlation in the Americas and among Sparkling wines.

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# Limitations & Areas for Further Research

## Limitations

- Preference bias
- Reputation and branding premium
- Unbalanced dataset

## Areas for Further Research

- Time-series analysis
- Consumer preferences by demographics
- Regional and Winery isolated studies

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# Thank You!

Any Questions?

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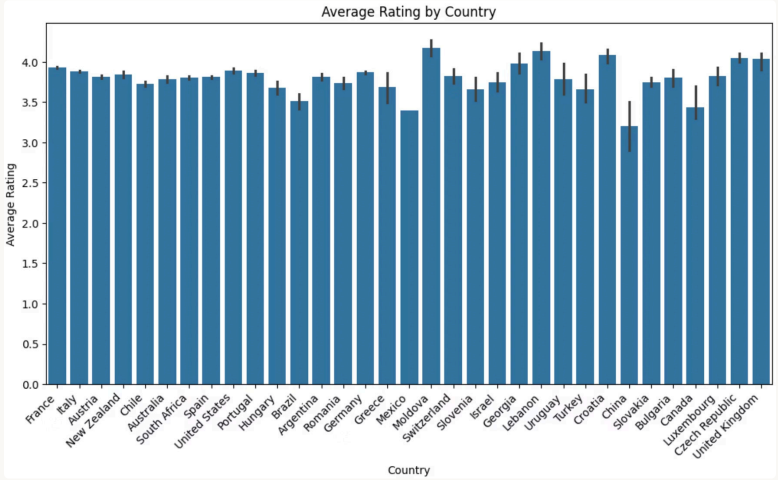
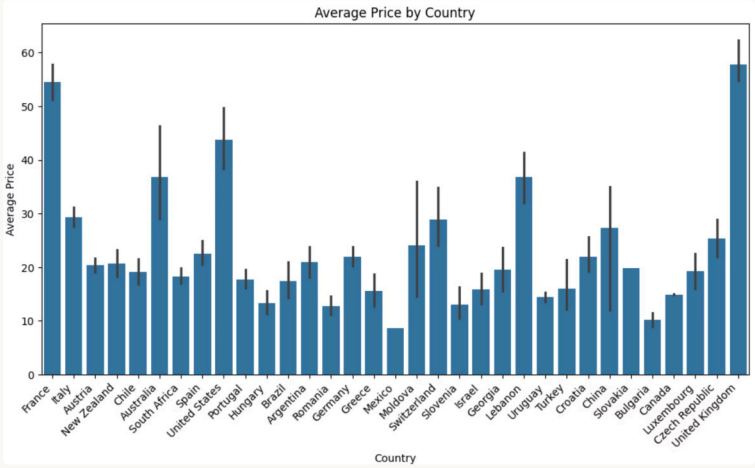
# Appendix

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